

Functional Features included in the solution

Date	07 July 2026
Project Name	CaptionAI: AI Powered Social Media Caption Generator
Maximum Marks	5 Marks

Functional Features Included in the Solution:

The table below provides a comprehensive list of all functional features successfully integrated into CaptionAI, showing the modules developed and the exact capabilities the application provides to end users.

S. No	Feature Name	Description & Purpose	Module / Layer	Status
1	User Registration	Allows new users to create an account by providing their name, email address, and password, to provide secure access and personalized user accounts. Users can successfully register and access the application.	Authentication Module	Completed
2	User Login	Registered users can securely log in using their email address and password to authenticate before accessing AI-powered features, ensuring only authenticated users can use the application.	Authentication Module	Completed
3	AI Caption Generation	Users enter a topic or prompt and the system generates engaging social media captions using the Groq API, reducing the time and effort required to create creative content. High-quality captions are generated within a few seconds.	Caption Generation Module / AI Processing Layer	Completed
4	Platform Selection	Users can select the target social media platform (Instagram, LinkedIn, Facebook, X, etc.) before generating captions, so that captions are suitable for different platforms with an appropriate writing style.	Frontend UI / Caption Generation Module	Completed
5	Tone Selection	Users can choose the preferred writing tone such as Professional, Casual, Friendly, Motivational, or Humorous, customizing captions according to their communication style so they match the selected tone and audience.	Frontend UI / Caption Generation Module	Completed
6	Hashtag Generation	The application automatically generates relevant hashtags along with the caption to improve content visibility and social media reach, without manual searching.	AI Processing Layer / Caption Generation Module	Completed
7	Call-to-Action (CTA) Suggestions	The system generates engaging call-to-action statements related to the generated caption to increase audience interaction, encouraging likes, comments, shares, or visits.	AI Processing Layer / Caption Generation Module	Completed

S. No	Feature Name	Description & Purpose	Module / Layer	Status
8	Caption History	All generated captions are stored in the SQLite database and can be accessed later, helping users reuse or review previously generated content whenever required.	History Module / Data Layer	Completed
9	Copy & Download	Users can copy generated captions or download them for future use, improving usability and providing quick access to generated captions for publishing or editing.	Download & Copy Module / Frontend UI	Completed
10	Secure Data Storage	User information and generated captions are stored securely in the SQLite database to maintain user records and application data safely, ensuring reliable data storage, retrieval, and improved application stability.	Data / Storage Layer	Completed

Functional Feature Summary

The CaptionAI application provides an integrated platform for AI-assisted social media content creation. It combines secure user authentication, AI-powered caption generation, hashtag and CTA suggestions, caption history management, and download functionality within a simple and user-friendly interface. These features work together to improve productivity, reduce manual effort, and support consistent content creation across multiple social media platforms.